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2026/2027

GitHub & Collaboration

Professional workflows, branch security, and continuous deployment.



Session 3 · with Soufiane

Course Plan

A step-by-step guide to mastering team collaboration.

01

Git Concepts Recall

02

Remotes & Cloning

03

Roles, Permissions & Security

04

Pull Requests & Merging

05

Sync & GitHub Pages

06

Demo & Practice

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Git Essentials Recall

S

Snapshots

Saving local progress permanently with descriptive messages.

I

Isolation

Working on parallel features without disrupting the core project.

R

Remote Sync

Uploading local history to a shared central server.

M

Integration

Combining finished parallel work back into the main timeline.

Git vs GitHub

Git

Tool

Version control that runs on your computer and tracks your project history locally.

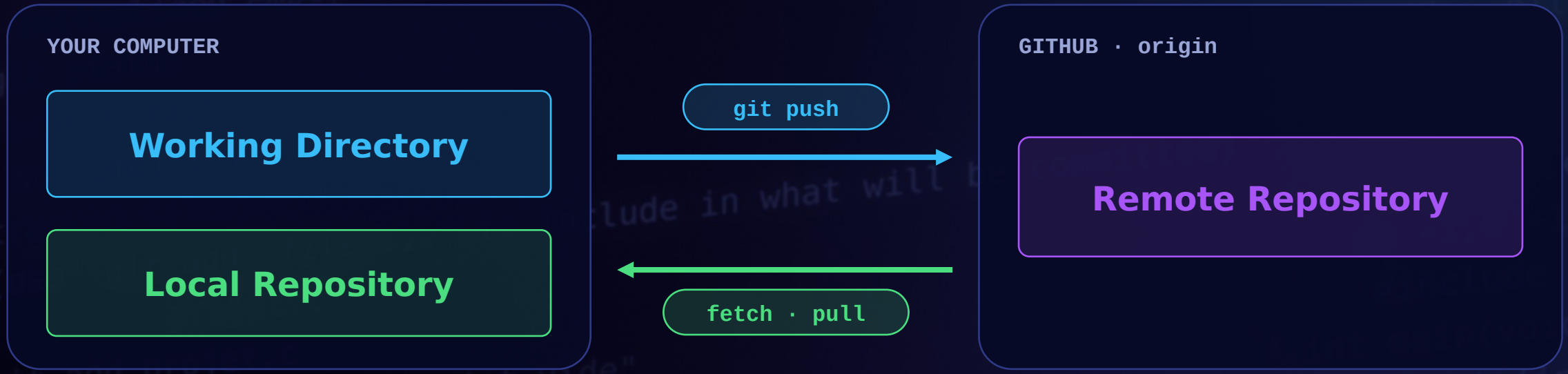
GitHub

Platform

Hosts your Git repositories online for teamwork: Pull Requests, Issues, Actions and Pages.

REMEMBER **Git is the tool. GitHub is the platform built around it.**

Local vs Remote Repository



`> git remote add origin <url>`

Link a remote

`> git push -u origin main`

Publish your commits

`> git clone <url>`

Copy a remote locally

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DEMO 1

Cloning the Repository

Getting the project onto your local machine.

- ✓ **Downloads complete project history.**
- ✓ **Creates an isolated local workspace.**
- ✓ **Automatically links to the remote server.**

```
> git clone <url>

> cd project

> git status
```

Actors and Permissions

O

Owner

Full repository control.

Manages settings, security rules, and team access.

C

Collaborator

Direct write access.

Can push to internal branches and merge standard code.

E

External Dev

No direct write access.

Must use Fork and Pull workflow to suggest changes.

The Fork & Pull Workflow

1

Fork

Copy the repo to your own account.

2

Clone

Download your fork locally.

3

Branch & Push

Commit on a feature branch and push it.

4

Open a PR

Propose your changes to the original repo.

5

Merge

The maintainer reviews and merges.

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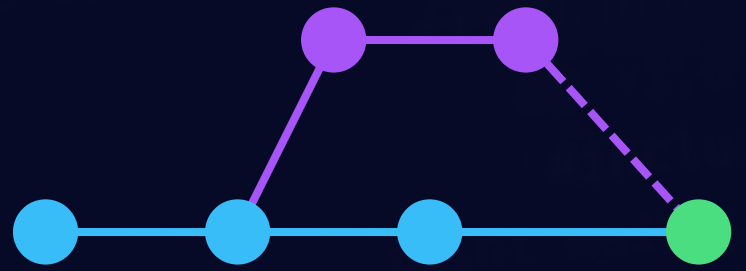
Protecting the Main Branch

Using Rulesets to enforce quality and prevent accidental deletions.

Strict Security Rules:

- ✕ Direct pushes heavily restricted.
- ✓ Pull Requests made mandatory.
- ● Minimum peer review approvals required.

`Settings → Rules → Rulesets → New branch ruleset`



main · protected

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DEMO 2

The Pull Request Cycle

1

Branch

Isolate new features locally away from production.

`git switch -c feature`

2

Push

Upload the feature branch to GitHub.

`git push -u origin feature`

3

Review

Team inspects code, requests changes, or approves.

`On GitHub`

4

Merge

Safely integrate approved work into the main branch.

`Merge button`

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Anatomy of a Good Pull Request

1

Clear title

Say what the change does in one line.

2

Useful description

Explain what changed and why.

3

Small scope

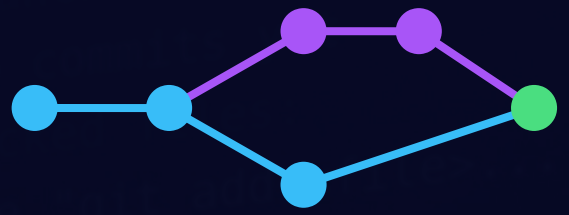
One feature or fix per Pull Request.

4

Respond to review

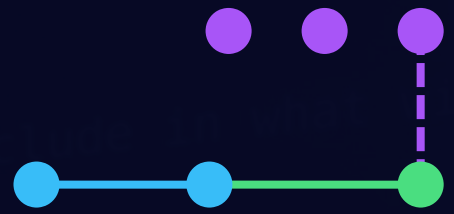
Push fixes to the same branch: the PR updates itself.

Three Ways to Merge a PR



Merge commit

Keeps every commit and adds a merge commit M.



Squash and merge

Combines all PR commits into one clean commit.



Rebase and merge

Replays commits on top of main for a linear history.

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DEMO 3

Local Synchronization Strategies

Fetch

`> git fetch`

- Downloads remote updates only.
- Leaves your local files completely untouched.
- Perfect for safely reviewing incoming work.

Pull

`> git pull`

- Downloads all remote updates.
- Immediately merges changes into local files.
- Can cause sudden conflict resolutions.

GitHub Pages Deployment

Turn your repository into a live website instantly.

- ✓ **Free native hosting infrastructure.**
- ✓ **Supports HTML, CSS, JS,React.js static files.**
- ✓ **Auto-updates upon main branch push.**

```
Settings → Pages → Branch: main
```



username.github.io

Your project live on the web.

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What Have We Learned?

Concept	Meaning
Remote	Repository hosted on GitHub (origin)
Clone	Copy a remote repo locally
Fork	Your own copy of another repo
Pull Request	Reviewed proposal to merge a branch
Protection	Rules that guard the main branch
Fetch / Pull	Download only / download + merge

COLLABORATION FLOW

Branch

Push

Pull Request

Review

Merge

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Demo

Let's build the collaborative repository together.



Let's Practice!

SET UP

1 Create a GitHub repository with a README

```
> github.com → New repository
```

2 Clone it to your computer

```
> git clone <url>
```

3 Create a branch and edit README.md

```
> git switch -c feature
```

COLLABORATE

4 Commit and push the branch

```
> git push -u origin feature
```

5 Open a Pull Request on GitHub

```
> Compare & pull request
```

6 Review, merge, then update locally

```
> git switch main · git pull
```

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Thank you for listening!

Join us again next week!!